PROBLEM STATEMENT #16 | Healthcare and Telemedicine

Remote Patient Vitals Alert & Monitoring App

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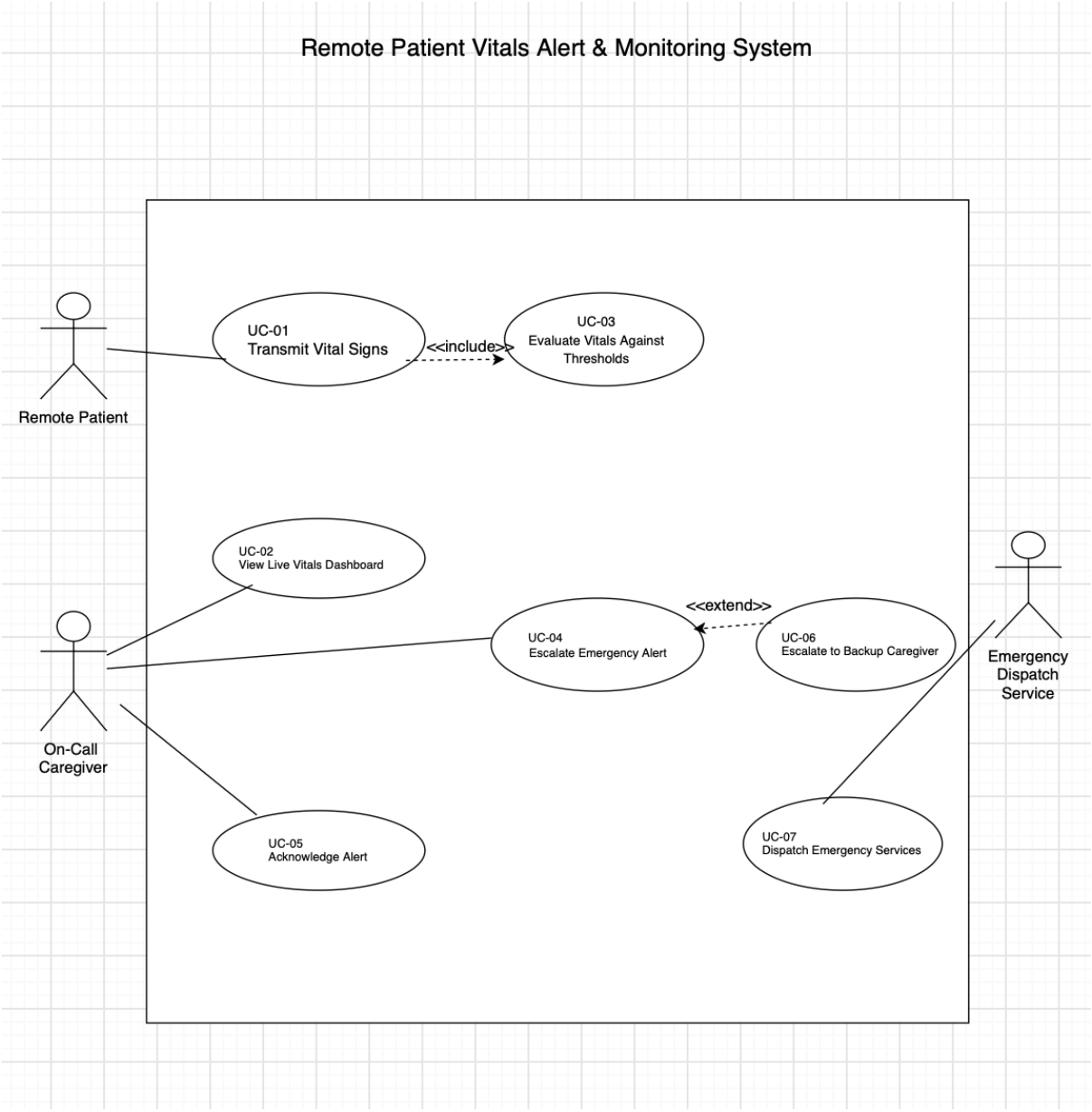
SECTION: G

1.Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall continuously evaluate ingested patient vitals against configured clinical thresholds and flag critical anomalies within 2 seconds.	High	Pass: Heart rate >140 BPM triggers immediate escalation. Fail: Alert delayed >5 seconds.	Quickly detects critical patient conditions.
FR-002	Functional	The system shall continuously collect and display the patient's SpO_2, heart rate, and blood pressure readings.	High	Pass: All received vital readings are displayed. Fail: Any reading is missing.	Enables continuous patient monitoring.
FR-003	Functional	The system shall send an emergency alert to the assigned on-call caregiver when a critical vital anomaly is detected.	High	Pass: Critical anomaly generates an alert. Fail: No alert is generated.	Ensures caregiver receives emergency notification.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-004	Functional	The system shall allow the on-call caregiver to view the patient's current and recent vital readings.	High	Pass: Caregiver can view current and recent readings. Fail: Readings cannot be accessed.	Helps caregiver assess patient condition.
FR-005	Functional	The system shall record detected anomalies and emergency alerts with patient information and timestamp.	Medium	Pass: Every anomaly and alert is recorded. Fail: Event is not recorded correctly.	Maintains history for monitoring and review.
NFR-001	Nonfunctional	The telemetry ingestion gateway shall support at least 500 concurrent continuous telemetry streams with 99.99% uptime.	High	Pass: System meets 500-stream and uptime targets. Fail: Target is not met.	Ensures reliability under high load.
NFR-002	Nonfunctional	The system shall restrict access to patient vital information to authenticated and authorized users.	High	Pass: Only authorized users can access data. Fail: Unauthorized access is possible.	Protects sensitive patient information.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: Escalate Emergency Alert

Use Case ID: UC-04
Use Case Name: Escalate Emergency Alert
Primary Actor: On-Call Caregiver
Supporting Actor: Remote Patient

Preconditions

1. The patient is registered in the Remote Patient Vitals Alert & Monitoring System.
2. The vital monitoring device is connected to the system.
3. Patient vital signs are being continuously transmitted to the system.
4. Clinical thresholds for patient vital signs have been configured.
5. An on-call caregiver has been assigned to the patient.
6. A critical vital anomaly has been detected by the system.

Postconditions

1. The emergency alert has been successfully generated.
2. The emergency alert has been sent to the on-call caregiver.
3. The on-call caregiver has been notified of the critical patient condition.
4. The alert status has been recorded in the system.
5. If the alert is not acknowledged, it has been escalated to the backup caregiver.

Main Success Scenario

1. The system detects a critical abnormality in the patient's vital signs.
2. The system identifies the abnormal vital sign and evaluates it against the configured clinical threshold.
3. The system creates an emergency alert containing the patient's information and abnormal vital reading.
4. The system sends the emergency alert to the assigned on-call caregiver.
5. The on-call caregiver receives the emergency alert.
6. The on-call caregiver views the patient's current vital readings.
7. The on-call caregiver reviews the emergency alert.
8. The on-call caregiver acknowledges the alert.

9. The system updates the alert status as acknowledged.
10. The system records the alert and acknowledgement with the date and time.
11. The use case ends successfully.

Alternate Flow

4a. On-Call Caregiver Does Not Acknowledge the Emergency Alert

1. The system detects that the emergency alert has not been acknowledged within the configured response time.
2. The system keeps the emergency alert active.
3. The system escalates the emergency alert to the backup caregiver.
4. The backup caregiver receives the escalated emergency alert.
5. The system records the escalation event with the date and time.
6. The backup caregiver reviews the emergency alert.
7. The use case ends after the alert has been escalated successfully.

Result

The system provides the On-Call Caregiver with an emergency alert containing the patient's critical vital information. If the alert is not acknowledged within the configured response time, the system escalates the alert to the backup caregiver and records the escalation event.